Congress is Non-binary: A Mixed Membership Exploration of the 118th Congress

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Abstract

PUT abstract here

1. Introduction

The two-party system has long been a hallmark of American politics, with the Democrats and Republicans serving as the most well-known parties, and many politicians firmly associating themselves with either or. But is the distinction between beliefs actually that clear? Can our politics really be reduced to a simple Democrat/Republican distinction? In this paper, we seek to explore whether or not politicians' voting behavior & congressional speeches adds another dimension to consider — hence, is Congress non-binary?

To start, we begin by inspecting whether or not there might be other communities at hand in the House, and how mixed representatives' memberships are. We note that members often don't belong to just one faction, but rather, sit between many of them. To do this, we conduct mixed-membership community detection by modeling House members with a degree-corrected mixed-membership (DCMM) model, in which each member has a membership vector π_i over K communities. We estimate π_i with Mixed-SCORE (Jin, Ke & Luo), a spectral method, extended with node covariates following Ji & Jin.

We then inspect the words that each community uses to analyze if there are any trends and notable distinctions between topics that they tend to talk about more on the House floor. To do this, we start with some basic natural language processing and cleaning, before reweighting so that each representative has equal weight in the process and no single representative skews their community's word frequencies. We then weight community word counts with soft membership, where words frequently stated by individuals that toe faction lines are weighted into their groups by what % of V_ a representative is.

INSERT CONCLUSION HERE

2. Methodology

2.1. Data

2.1.1. Speeches

Floor speeches were sourced from the daily edition of the Congressional Record, the official and publicly available account of the proceedings and debate of the United States Congress, published by the Government Publishing Office (GovInfo). They were extracted with downloading and parsing code adapted from the boundaries_congress package (van der Does and Kolchinsky, 2021), which Zhao (2025) also used. Speeches were collected for the 112th through the 119th Congresses. The speech features cover the 115th, 116th, and 118th Congresses; the 117th is set aside because redistricting affects the baseline, and the 119th is incomplete.

Zhao's pipeline filtered speeches heavily: procedural speakers were removed, speeches had to be at least 250 characters long and have a document title containing "Act" or "Resolution," and the remaining speeches were matched to bills by title. In the 117th Congress this retained 10,485 of about 96,747 speeches (about 11%). A further coverage filter on speakers and topics left 134 legislators, and the thesis notes that this selection may overrepresent members of leadership and committee chairs.

Because the present analysis models the whole House, a lighter selection was used:

- House speeches with a member identifier (bioguide ID)
- At least 250 characters
- Dated before November 1 of the election year, which avoids using information from after the election

In the 118th Congress, this selection reduced 97,817 speeches (House and Senate) to 28,914. Among the 320 incumbents in the 118th modeling sample, 319 have at least five retained speeches and 280 (88%) have at least twenty, with a median of 47.5 speeches per member. Participation in floor speech remains uneven, so some selection toward more

active speakers is expected in the member-level speech features. About one third of all rows have no member identifier, mostly presiding officers and similar roles. The Records for January 3 of 2023 (the opening day of the 118th Congress) is unavailable from GovInfo.

2.1.2. VOTING RECORD

To generate a graph which represents voting information, we first pulled the voting records from the Voteview database. This allowed us to attain a record of representatives and how they voted in a congress. We focus on the 118th Congress and use only final passage roll calls on bills and joint resolutions (360 votes).

We then built an undirected weighted graph where two representatives are connected with a weight corresponding to how similarly they voted. For instance if two congress people cast the same vote on x number of votes and cast different votes on y votes they would be connected with a weight of $S_{ij} = \frac{x}{x+y}$. Here only votes on which both representatives cast a Yea or Nay vote are counted with absences ignored. Every pair of representatives is connected with no self connection. We call this matrix, S, the agreement matrix.

We require representatives to have voted on at least half of the bills (180 of 360) to be included, to avoid noisy and poorly behaved nodes, since a representative with few votes has agreement weights based on only a few shared bills. This leaves 428 of the 445 representatives who served in the 118th Congress. The adjacency weight matrix is included as figure (X COME BACK TO AND PUT THE NUMBER IN) in the appendix.

2.2. Mixed Membership Model

We first decided on how many communities to try and detect. To do this we looked at the absolute values of the eigenvalues of the agreement matrix. Looking at the scree plot (figure X_2 COME BACK TO AND ADD FIGURE NUMBERS) we see that after the fourth eigenvalue the other eigenvalues level off and do not appear to carry significant signal. We therefore decided to run a mixed membership community detection with $K = 4$ communities.

We use our weighted matrix S in place of A in the Mixed-SCORE algorithm for the DCMM model, where $\Omega_{ij} = \theta_i\theta_j\pi_i^\top P\pi_j$ and $E[A] = \Omega - \mathrm{diag}(\Omega)$. Since S_{ij} is the frequency with which congress person i and j vote the same way, it is an estimate of $P(i$ votes the same as $j)$, and we treat it as an approximation of $E[A_{ij}] = \Omega_{ij}$. This means that S is a nearly noiseless approximation of $E[A]$. Let $\hat{\lambda}_k$ and $\hat{\xi}_k$ denote the k-th eigenvalue and eigenvector of S, ordered by absolute value. Taking the 4 leading eigenvectors of S, we performed the SCORE algorithm, which with $K = 4$ gives

$$\hat{R} = \left[\frac{\hat{\xi}_2}{\hat{\xi}_1}, \quad \frac{\hat{\xi}_3}{\hat{\xi}_1}, \quad \frac{\hat{\xi}_4}{\hat{\xi}_1}\right]$$

We then added in embeddings of speeches each congressional member had made as covariate information. After standardizing the embedding's 5 PCs, we took the top $K = 4$ left singular vectors, call these $\hat{\eta}_1, \hat{\eta}_2, \hat{\eta}_3, \hat{\eta}_4$, and define

$$\hat{H} = \left[\hat{\eta}_1, \quad \hat{\eta}_2, \quad \hat{\eta}_3, \quad \hat{\eta}_4\right].$$

We then rescaled $\hat{H}$ so its total variability across members (sum of squared robust standard deviations of columns) is the same as that of $\hat{R}$. We used the coefficient $c_{\text{scale}} = 13.9$. Then we used $\tau = \tau_0 \cdot c_{\text{scale}}$, where τ_0 is the relative weight tuning parameter. We used $\tau_0 = 1$ (equal weight), giving $\tau = 13.9$. This gives

$$Z = \left[\hat{R},\ \tau\hat{H}\right].$$

Once we have Z, we use the sketched vertex search, running k-means with $10 \cdot K = 40$ clusters and then applying the successive projection algorithm to the cluster centers to find the K vertices $\hat{v}_1, \ldots, \hat{v}_K$. We then expressed each member as a convex combination of these vertices using barycentric weighting, solving for each member i

$$\hat{w}_i = \arg\min_{w\geq 0,\ \sum_k w_k = 1} \left\| z_i - \sum_{k=1}^{K} w_k \hat{v}_k \right\|^2,$$

where z_i is the i-th row of Z. Finally we rescaled these weights to account for the degree parameters, $\hat{\pi}_i^*(k) = \hat{w}_i(k)/\hat{b}_1(k)$ with $\hat{b}_1(k) = \left[\hat{\lambda}_1 + \hat{v}_k^{R\top}\mathrm{diag}(\hat{\lambda}_2, \ldots, \hat{\lambda}_K)\hat{v}_k^R\right]^{-1/2}$, where $\hat{v}_k^R$ is the first $K-1$ coordinates of $\hat{v}_k$ (the part corresponding to $\hat{R}$), and normalized to obtain $\hat{\pi}_i = \frac{\hat{\pi}_i^*}{\|\hat{\pi}_i^*\|_1}$ so each member's weights sum to one.

2.3. WORLD CLOUDS (RENAME THIS SECTION, DOES THIS BELONG IN METHODOLOGY?)

3. Results

RESULTS

-TO do:

- Word Clouds

- Subcaucues

4. Discussion

DISCUSSION

5. Conclusion

Conclusion

6. Appendix

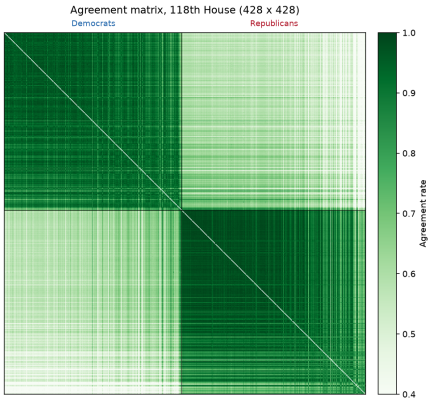


Figure 1. Agreement Matrix S

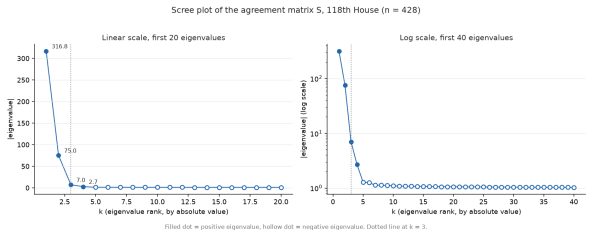


Figure 2. Scree plot showing magnitude of eigenvalues of S

7. Code Availability

Code used in this project can be found in this Google Drive link:

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